Circular economy

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1. Circular economy



Image: towardszerowaste.sg



Recover and regenerate products and materials at the end of each service life.

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1. Circular economy

❖ One Definition of Circular Economy

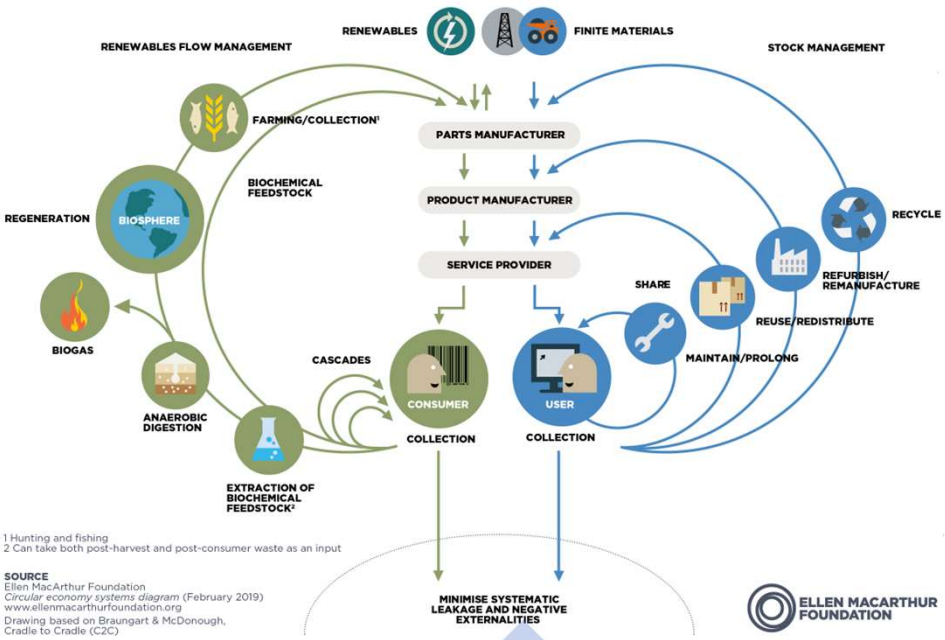
A circular economy describes an economic system that is based on business models which replace the "end-of-life" concept with reducing, alternatively reusing, recycling and recovering materials in production/distribution and consumption processes, thus operating at the micro level (products, companies, consumers), meso level (eco-industrial parks) and macro level (city region, nation and beyond), with the aim to accomplish sustainable development, which implies creating environmental quality, economic prosperity and social equity, to the benefit of current and future generations.

(Kirchherr et al., 2017)

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The circular economy is based on three principles, driven by design:

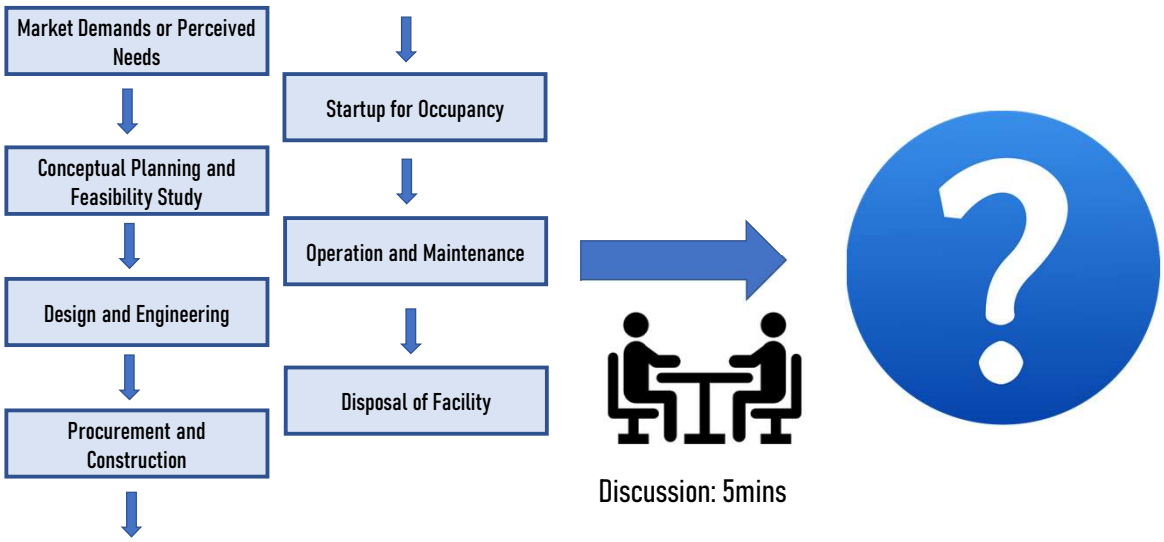
- Eliminate waste and pollution
- Circulate products and materials (at their highest value)
- Regenerate nature



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1. Circular economy



Discussion: 5mins

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1. Circular economy

❖ 9 R

	Strategies	
Smarter product use and manufacture	R0 Refuse	Make product redundant by abandoning its function or by offering the same function with a radically different product
	R1 Rethink	Make product use more intensive (e.g. by sharing product)
	R2 Reduce	Increase efficiency in product manufacture or use by consuming fewer natural resources and materials
Extend lifespan of product and its parts	R3 Reuse	Reuse by another consumer of discarded product which is still in good condition and fulfils its original function
	R4 Repair	Repair and maintenance of defective product so it can be used with its original function
	R5 Refurbish	Restore an old product and bring it up to date
	R6 Remanufacture	Use parts of discarded product in a new product with the same function
	R7 Repurpose	Use discarded product or its parts in a new product with a different function
Useful application of materials	R8 Recycle	Process materials to obtain the same (high grade) or lower (low grade) quality
	R9 Recover	Incineration of material with energy recovery

(Potting et al., 2017)

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2. Implications for the industry: Scale of opportunity

❖ **Reuse**

By 2040, needed investment in global infrastructure $94 trillion

Roads sector $30-$35 trillion

Assuming the average cost saving through Reuse 5%-15%

With every 1% of the road sector using…, the potential market is $15-$53 billion.

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2. Implications for the industry: Scale of opportunity

❖ **Recycle**

Construction and demolition waste: 374 million tonnes in the EU-28

70% recovery target of 2020, most achieved through backfilling and low-grade recovery applications.

The potential for recycling is still huge when high-grade recovery techniques are available. C2CA technology is among one of them.

❖ **Repair**

SHM: $2.9 billion by 2025; routine bridge inspections: $6.3 billion by 2029

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3. Reuse - 1



1.6km away



Original location of the building - Leigh Road, Slough

Plot for relocation of the building - 9 Cambridge Avenue, Slough

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3. Reuse - 1





Reuse - 1

- Original location (Leigh Road) – brick cladding New location (Cambridge Avenue)

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3. Reuse - 1



Schematic of the relocated building (Sturgis carbon profiling)

The following elements were successfully recovered and reused:

- Steel frame and balustrading
- Precast concrete ground beams
- Precast beams and planks (offices)
- Raised floor
- Staircases
- Curtain walling and glazing
- Fascia
- Roller doors
- Internal joinery
- Fencing and gates
- Loading doors
- Lift

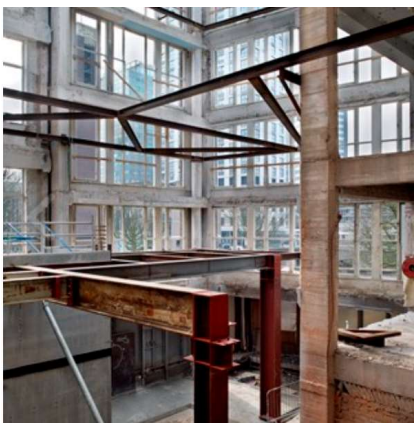
56% Embodied Carbon saving
25% Cost saving
6% lower whole life carbon footprint
No organic waste

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3. Reuse - 2



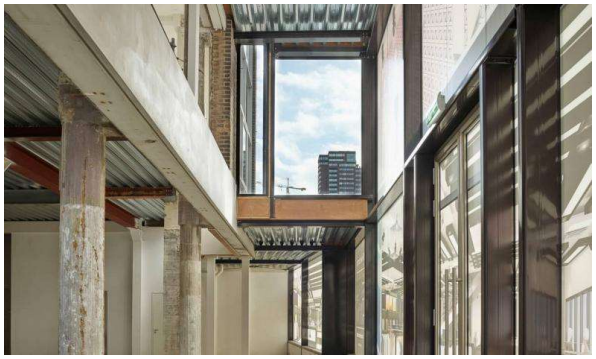


Winkelpanden Hoogstraat 168-172, Rotterdam

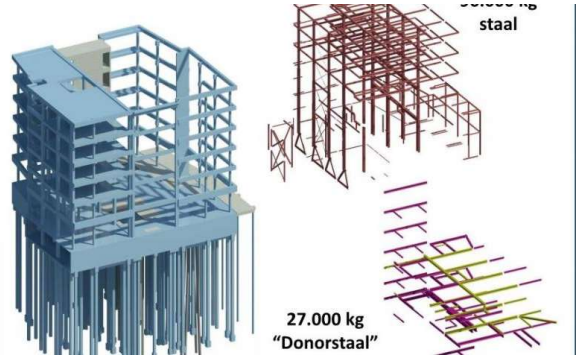
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3. Reuse - 2



Concrete floors are demolished.

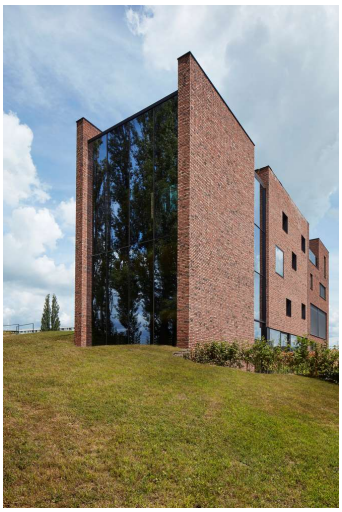


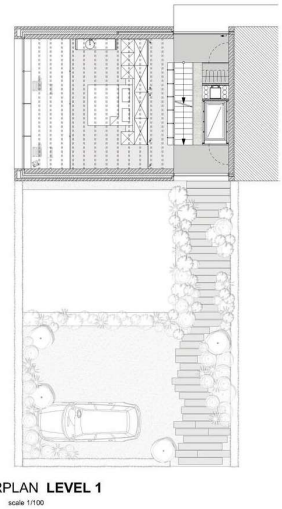
Total amount of reused steel.

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3. Reuse - 3



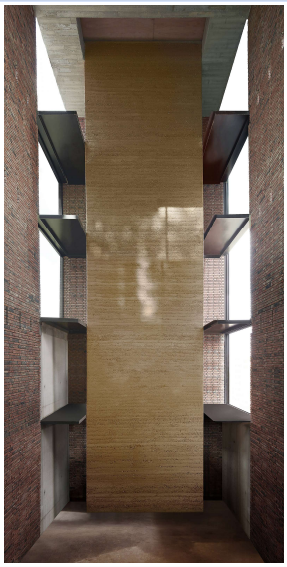


A house with rammed earth wall in Belgium-Circular Brick House

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3. Reuse - 3



The rammed earth wall 40 centimeters thick and 15 meters high, as the core of the house.

Advantages
- Same Compressive strength as concrete;
- 100% reusable material.

Disadvantages
- Weak tensile strength due to lack of steel rebars;
- Not waterproof.

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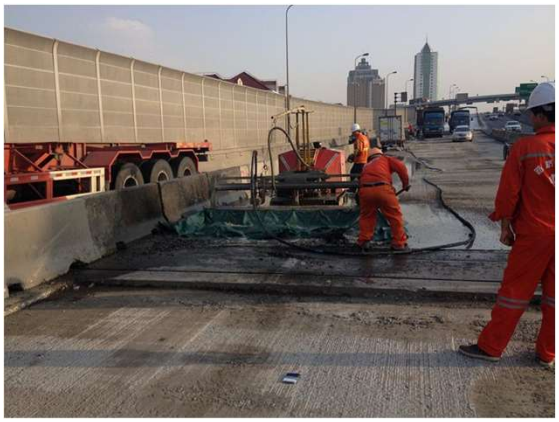
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4. Repair

Repair the cracks on the beam



Replace concrete deck on the bridge



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5. Rethink

❖ **Operating profit optimization task**

Operating Profit — Revenue — Cost

$$\max_{\boldsymbol{x}} \; P(\boldsymbol{x}) = \sum_{t=t_0}^{t_0+\Delta t} \left[\sum_{k\in\mathcal{K}} x_k \cdot v_k(x_k) \right] - \left[M\delta_M(t) - H\delta_H(t) - OP \right] - CC$$

$$s.t.$$

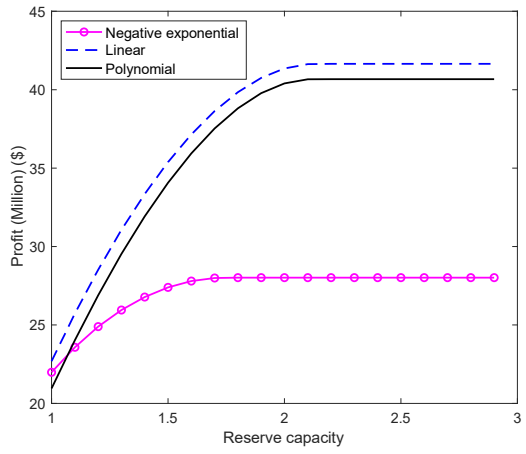
$$(a) \quad \boldsymbol{x}_{min} \le \boldsymbol{x} \le \boldsymbol{x}_{max}, \boldsymbol{x} \in \mathbb{R}^{|\mathcal{K}|}$$ → Constraint of toll rates

$$(b) \quad \sum_{k\in\mathcal{K}} v_k \cdot \alpha_k \le y$$ → Constraint of traffic volume

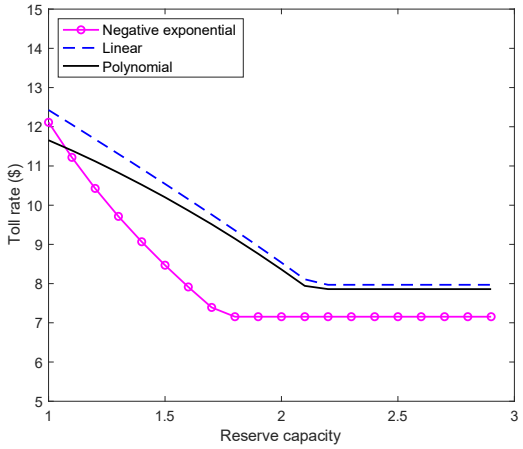
$$(c) \quad W(\boldsymbol{v}) \le RCap$$ → **Safety constraint**

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5. Rethink



Reserve capacity vs Profit



Reserve capacity vs Toll rates

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6. Repurpose

From Ship manufacturing to a Culture and cuisine place



Schiffbau, Zurich

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7. Flexible design



The Leeuwarden Energy Knowledge Centre

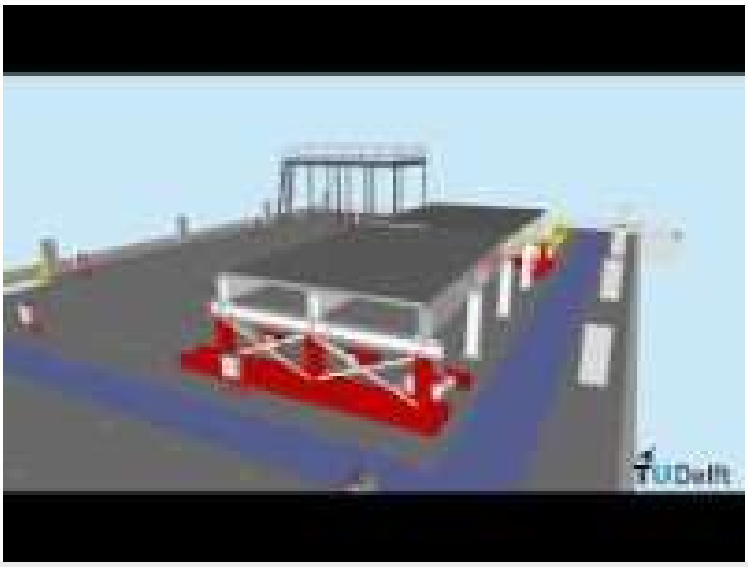


Adjustable steel support of the building

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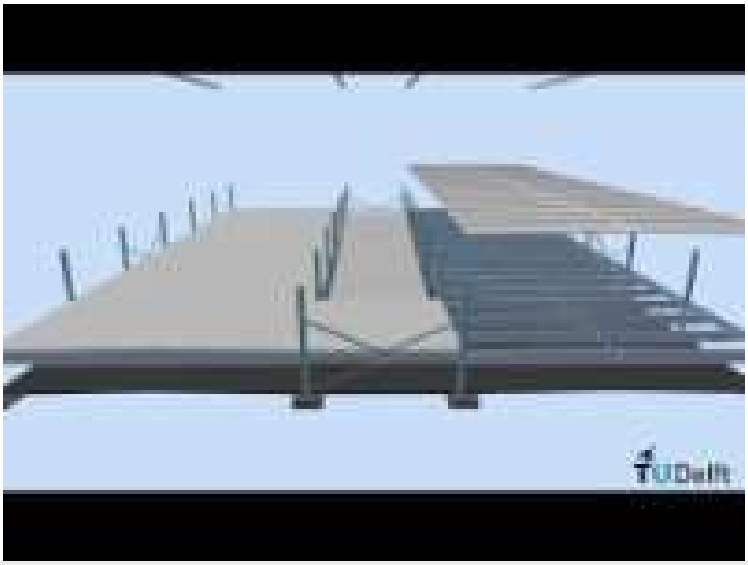
8. Design for Disassembly and Deconstruction



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8. Design for Disassembly and Deconstruction



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8. Design for Disassembly and Deconstruction

- Principles:
  - Use a simple, modular design
  - Use an open, flexible building system that allows the functions to change in the future
  - Use a modular structural grid
  - Design building so elements are layered according to their anticipated lifespan
  - Make sure stability is maintained during deconstruction
  - Separate mechanical, electrical and plumbing system

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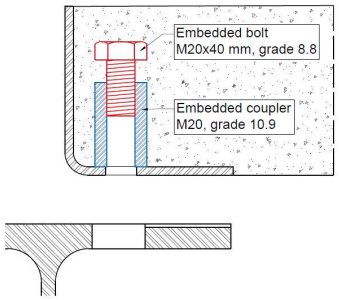
8. Design for Disassembly and Deconstruction

- Materials and connections:
  - Minimize the number of different materials, connections and components
  - Design joints that are accessible and durable
  - Use mechanical joints (bolts, nuts) instead of other types of joints
  - Use nontoxic, noncomposite, durable, and high-quality materials that can be reused
  - Avoid use of binders, adhesive, resin, and secondary finishes
  - Use recycled and recyclable materials.
  - Use lightweight materials.

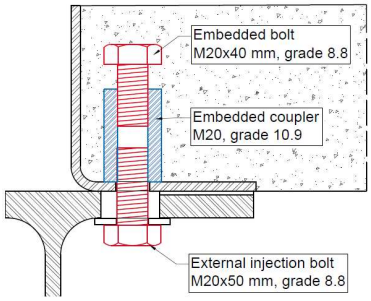
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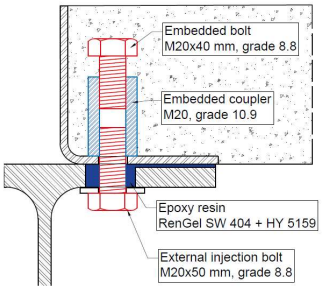
8. Design for Disassembly and Deconstruction



Lifting the concrete floor element over the steel beam



Aligning the embedded coupler in the steel beam



Injecting the bolt-hole clearance with the epoxy resin system

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8. Design for Disassembly and Deconstruction

- Construction and Deconstruction Phase
  - Develop and design a deconstruction plan already in the design process
  - Use prefabricated components and materials
  - Make sure components are sized to suit handling
  - Deconstruction should be possible with common tools and equipment.
  - Allow for parallel disassembly
  - Ensure access to building components

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8. Design for Disassembly and Deconstruction

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- Communication, Competence, and Knowledge
  - Information, documentation about used materials, and deconstruction method (and as-built drawings) need to be stored.
  - Component types should be identifiable.
  - Material types should be identifiable.

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8. Design for Disassembly and Deconstruction

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Modular bridge with precast sections designed to be disassembled and reuse.

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8. Design for Disassembly and Deconstruction



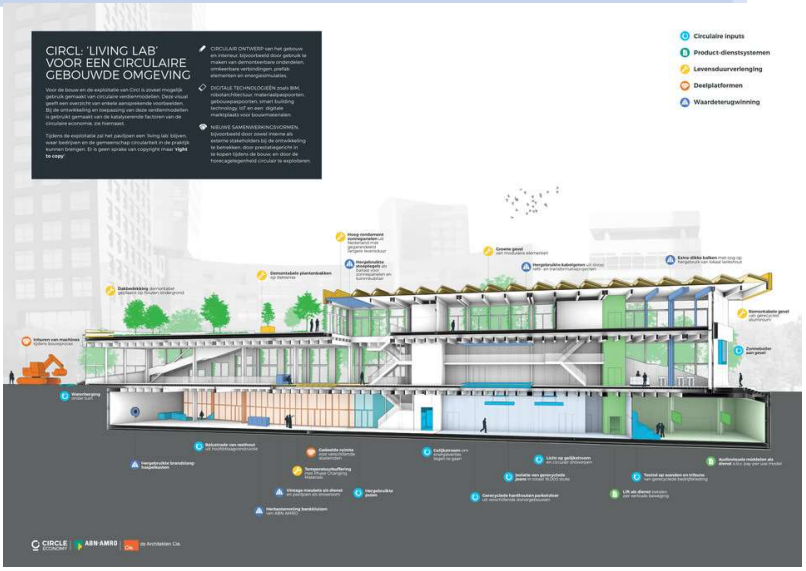


Pavilion Circl, Amsterdam

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8. Design for Disassembly and Deconstruction



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